

Multiple Choice (10 marks)

1. According to the Solar Nebular theory what are asteroids and comets?
 - (a) They are the shattered remains of collisions between planets.
 - (b) They are chunks of rock or ice that condensed long after the planets and moons had formed.
 - (c) They are chunks of rock or ice that were expelled from planets by volcanoes.
 - (d) They are leftover planetesimals that never accreted into planets.
2. The so-called "bigfoot" on Mars was actually a rock that was about 5 cm tall. It had an angular size of about 0.5 degrees (~30 pixels). How far away was this...
 - (a) About 6 meters
 - (b) About 6 feet
 - (c) About 10 meters
 - (d) About 10 feet
3. Which of the following is/are true?
 - (a) Titan is the only outer solar system moon with a thick atmosphere
 - (b) Titan is the only outer solar system moon with evidence for recent geologic activity
 - (c) Titan's atmosphere is composed mostly of hydrocarbons
 - (d) A and D
4. Which of the following most likely explains why Venus does not have a strong magnetic field?
 - (a) Its rotation is too slow.
 - (b) It has too thick an atmosphere.
 - (c) It is too large.
 - (d) It does not have a metallic core.
5. Which of the following was not cited as evidence for life in the martian meteorite ALH84001?
 - (a) Complex organic molecules specifically PAHs
 - (b) Magnetite grains similar to those formed by bacteria on Earth
 - (c) Carbonate minerals indicating a thicker warmer Martian atmosphere
 - (d) Amino acids with a preferred orientation or "chirality"

True / False (10 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'The axis of the Earth is tilted at an angle of approximately ... relative to the orbital plane around the Sun.' is '20.3 degrees'.
2. True or False: The correct answer to 'Which of the following is/are common feature(s) of all fresh (i.e. not eroded) impact craters formed on solid surfaces:' is 'A and B only'.
3. True or False: The correct answer to 'Most rocks on the Moon's surface are older than those on the Earth's surface. The best evidence for this is:' is 'The Moon's surface is more heavily eroded than the Earth's surface.'.
4. True or False: The correct answer to 'The constellation ... is a bright W-shaped constellation in the northern sky.' is 'Cassiopeia'.
5. True or False: The correct answer to 'One astronomical unit (AU) is equal to approximately ...' is '150 million km'.

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. Why do we look for water-ice in craters at Mercury's pole?
2. By locating the north celestial pole (NCP) in the sky how can you determine your latitude?

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